An ICT System and its components

Specification

Topic	Key Concepts	Content and Amplification
1. An ICT system and its components	What is ICT?	The use of technology for the input, storage, processing and transfer of data and the output of information.
	What is a system?	Any system involves input, processing and output.
	What is an ICT system?	ICT systems are those where the output from the system goes directly to a human being or into another ICT system.
	Any ICT system is made up of components.	The components are: • people • data • procedures • software • hardware • information.

To consolidate you should be able to answer these questions…

1. What is ICT?
2. What is a system?
3. What is an ICT System?
4. Define input
5. Define process
6. Define output
7. What does an IPO diagram do?

What is ICT

- Use of technology for **input**, **process**, storage and transfer of **data** and the **output** of **information**

What is a system?

- Any activity that involves input, processing and output
- E.g. Digestive system

What is an ICT System?

- A system that uses any form of digital technology, the output goes directly to a human being or into another systems
- A system that uses any form of digital technology to input and process data and out put the information which goes directly to a human or into another system.
- E.g. online holiday booking system

INPUT

- Entering data into the computer or other device

- Examples:
  - PIN number
  - Barcode
  - Phone numbers on a mobile phone

PROCESS

- Manipulating (calculating, querying, searching) the input data into information that is understandable to the user.

- Example:
  - Identifying customer account and balance in relation to PIN
  - Counting the number of items purchased and adding up the amount of money owed
  - Searching the contacts in your mobile phone, identifying the correct entry and activating a call

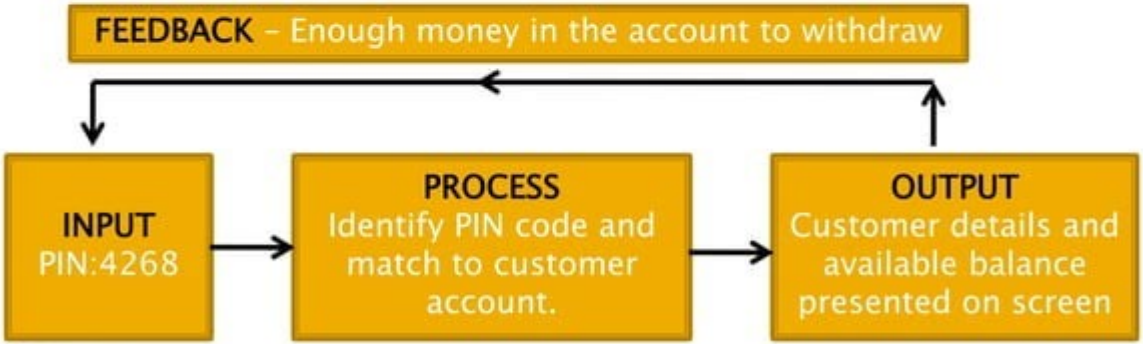
OUTPUT

- Presenting the information to the user, it must have a context. It could be printed, displayed, heard or in another form.

- Example:
- Onscreen presentation of available balance or printed output
- Receipt printed for items purchased, showing individual item costs and total price
- Dialing tone initiated

Input Process Output (IPO)

- **Input** is capturing the data, **process** is converting the data into information and **output** is the information produced.
- **Feedback** might then be used to affect another input (into the same system or a different system)
- IPO diagrams can be drawn to illustrate this process in a system



To consolidate you should be able to answer these questions…

1. What are the 6 components of an ICT system?
2. Define Data and give an example
3. Define Information and give an example
4. Define Hardware
5. Draw a diagram that illustrates parts of a computer system
6. Define software
7. Define procedures and give an example

What are the 6 components of an ICT system?

- An example of an ICT System is an Management Information System (MIS).
- Large organisations use these systems at a strategic level (Principles, Managing Directors, Leaders of a business)
- They use them to help collate and analyse data from other systems e.g. HR Systems, Transaction Processing Systems

What are the 6 components of an ICT system?



MIS

HR System



- Username and Password required to login to HR System
- Must comply with DPA
- Data backed up off site

Electronic Register



- Username and Password required to login to E1
- Presence and absence must be filled in by 8:15am
- Lates must be labelled with L
- Data is backed up on remote server

What are the 6 components of an ICT system?

People
Software
Hardware
Data
Information
Procedures



MIS



HR System



- Username and Password required to login to HR System
- Must comply with DPA
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Electronic Register



- Username and Password required to login to E1
- Register must be filled in by 8:15am
- Lates must be labelled with L
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What is Data?

- Data is raw facts and figures or a set of values, data has no meaning.
- Examination Results (DATA) : 3312, 2, 35, 3, 64, 4421, 2, 48, 3, 56

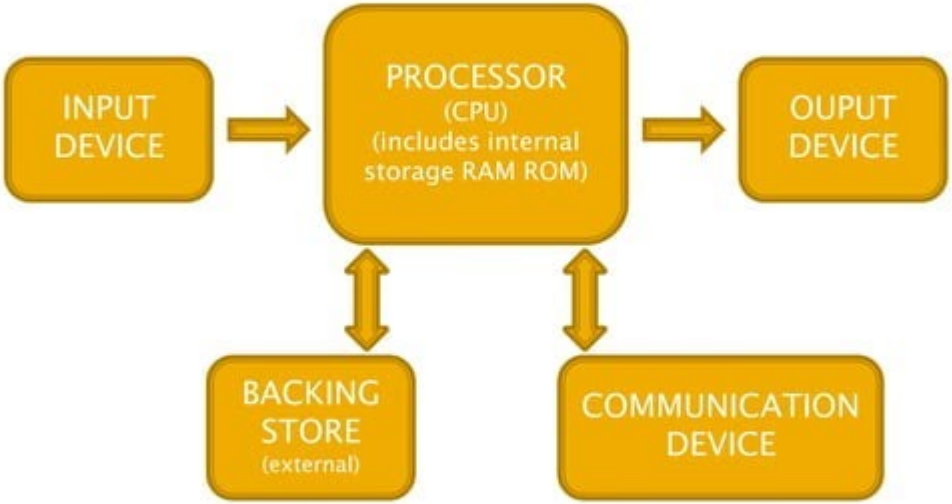
What is information?

- Information is data that has been processed and put into context to give it meaning
- Examination Results (DATA) : 3312, 2, 35, 3, 64, 4421, 2, 48, 3, 56
- The candidate number has been processed, the student found. The course code has been processed and matched to the student, the score for each course assigned to that student. This information can then be output in the form of a report or graph.

What is Hardware?

- Physical components of an ICT system e.g. Printer, scanner, monitor, server, switches
- In many systems data is transmitted from one computer to another.

Illustration of Hardware



What is software?

- Computer Programs that make the hardware do something useful

What are Procedures?

- A series of actions or operations intended to achieve a result
- ICT Systems usually involve a number of procedures that have to be undertaken to make sure that the system can run smoothly.
- Examples
  - A code of practice (guidelines and regulations set out by the company)
  - Legal procedures (DPA, Computer Misuse Act)
  - Backup procedures
  - Security procedures
  - Procedure to archive data
  - Specific tasks generate procedures also

People

- Testing of a system
- Sales
- Purchasing
- Finance
- Operations
- Internal and External personnel

Activity

- Try drawing an IPO diagram that might be used in an ICT System.

OR

- Choose a piece of software with which you are familiar with and draw an IPO diagram for a task you would use that software to perform